1. A rigid body is undergoing pure translation if every straight line in the body:

(A) Remains strictly parallel to its initial direction
(B) Rotates at a constant angular velocity
(C) Passes through the instantaneous center
(D) Experiences constant centripetal acceleration

2. During pure translation of a rigid body, the velocity and acceleration vectors of any two points A and B are:

(A) Identical at all instants (v_A = v_B, and a_A = a_B)
(B) Perpendicular to each other
(C) Inversely proportional to distance
(D) Zero

3. In fixed-axis rotation of a rigid body, all particles move in concentric circular paths whose centers lie on:

(A) The fixed axis of rotation
(B) The centroid of the body
(C) The instantaneous center
(D) The line of action of external torque

4. The linear tangential velocity v of a particle rotating at distance r from a fixed axis with angular velocity omega is:

(A) v = r * omega
(B) v = r * alpha
(C) v = omega / r
(D) v = r^2 * omega

5. In fixed-axis rotation, the tangential acceleration component a_t of a particle at radius r is:

(A) a_t = r * alpha
(B) a_t = r * omega^2
(C) a_t = alpha / r
(D) a_t = 0

6. In fixed-axis rotation, the normal (centripetal) acceleration component a_n directed toward the axis is:

(A) a_n = r * omega^2 = v^2 / r
(B) a_n = r * alpha
(C) a_n = v * alpha
(D) a_n = omega / r

7. The total linear acceleration magnitude 'a' of a particle rotating about a fixed axis is:

(A) a = r * sqrt(alpha^2 + omega^4)
(B) a = r * (alpha + omega^2)
(C) a = r * alpha
(D) a = r * omega^2

8. General plane motion of a rigid body can always be considered as the superposition of:

(A) A translation of an arbitrary reference base point and a relative rotation about that base point
(B) Two independent pure rotations
(C) Two collinear translations
(D) A translation and a dilation

9. The relative velocity equation between two points A and B on a rigid body undergoing planar motion is:

(A) v_B = v_A + v_B/A = v_A + (omega x r_B/A)
(B) v_B = v_A - v_B/A
(C) v_B = v_A * omega
(D) v_B = v_A + a_B * t

10. The relative velocity vector v_B/A of point B with respect to base point A is always:

(A) Perpendicular to the line connecting points A and B
(B) Parallel to line AB
(C) Directed toward point A
(D) Zero

11. The magnitude of relative velocity v_B/A is calculated as:

(A) |v_B/A| = omega * r_B/A
(B) |v_B/A| = alpha * r_B/A
(C) |v_B/A| = omega^2 * r_B/A
(D) |v_B/A| = v_A / r_B/A

12. The Instantaneous Centre of Zero Velocity (ICR or ICZ) is defined as a point in the plane of motion that has:

(A) Zero instantaneous linear velocity (v_ICR = 0)
(B) Zero instantaneous acceleration
(C) Zero angular velocity
(D) Infinite acceleration

13. At any given instant, the velocity of ANY point P on a rigid body is related to the ICR by:

(A) v_P = omega * r_P/ICR directed perpendicular to line P-ICR
(B) v_P = alpha * r_P/ICR
(C) v_P = v_ICR + r_P
(D) v_P = omega^2 * r_P

14. For a wheel or cylinder rolling WITHOUT SLIPPING on a stationary flat or curved surface, the ICR is located at:

(A) The instantaneous point of contact with the ground surface
(B) The geometric center of the wheel
(C) The highest point of the wheel
(D) Infinity

15. In a wheel of radius R rolling without slipping at angular velocity omega, the linear velocity of its center of mass G is:

(A) v_G = R * omega
(B) v_G = 2 * R * omega
(C) v_G = 0
(D) v_G = 0.5 * R * omega

16. In the wheel of question 15 rolling without slipping, the instantaneous linear velocity of the highest point (top) is:

(A) 2 * R * omega
(B) R * omega
(C) 0
(D) 4 * R * omega

17. In the wheel of question 15 rolling without slipping, the instantaneous linear velocity of the lowest contact point is:

(A) 0 (Zero)
(B) R * omega
(C) 2 * R * omega
(D) - R * omega

18. If the directions of velocities v_A and v_B at two distinct points A and B are known and non-parallel, the ICR is located at:

(A) The intersection of perpendicular lines drawn to v_A and v_B at points A and B
(B) The midpoint of line AB
(C) The intersection of velocity vectors v_A and v_B
(D) The center of mass

19. Mass moment of inertia I of a rigid body about an axis is defined by the volume integral:

(A) I = Integral(r^2 dm)
(B) I = Integral(r dm)
(C) I = Integral(r^3 dm)
(D) I = m * r

20. In SI units, the mass moment of inertia is expressed in:

(A) kg-m^2
(B) N-m
(C) kg/m^2
(D) kg-m/s

21. The mass moment of inertia of a slender uniform rod of mass m and length L about a transverse centroidal axis is:

(A) I_G = (1/12) * m * L^2
(B) I_G = (1/3) * m * L^2
(C) I_G = (1/2) * m * L^2
(D) I_G = m * L^2

22. The mass moment of inertia of a slender uniform rod of mass m and length L about a transverse axis through one end is:

(A) I_end = (1/3) * m * L^2
(B) I_end = (1/12) * m * L^2
(C) I_end = (1/2) * m * L^2
(D) I_end = (2/3) * m * L^2

23. The mass moment of inertia of a thin circular hoop or ring of mass m and radius R about its central polar axis is:

(A) I_G = m * R^2
(B) I_G = 0.5 * m * R^2
(C) I_G = 0.25 * m * R^2
(D) I_G = (2/5) * m * R^2

24. The mass moment of inertia of a solid uniform circular cylinder or disk of mass m and radius R about its polar axis is:

(A) I_G = 0.5 * m * R^2
(B) I_G = m * R^2
(C) I_G = 0.25 * m * R^2
(D) I_G = (2/5) * m * R^2

25. The mass moment of inertia of a solid uniform sphere of mass m and radius R about any centroidal diameter is:

(A) I_G = (2/5) * m * R^2
(B) I_G = (2/3) * m * R^2
(C) I_G = 0.5 * m * R^2
(D) I_G = m * R^2

26. The mass moment of inertia of a thin hollow spherical shell of mass m and radius R about any diameter is:

(A) I_G = (2/3) * m * R^2
(B) I_G = (2/5) * m * R^2
(C) I_G = 0.5 * m * R^2
(D) I_G = m * R^2

27. The Parallel Axis Theorem for mass moment of inertia is mathematically expressed as:

(A) I = I_G + m * d^2
(B) I = I_G - m * d^2
(C) I = I_G + m * d
(D) I = I_G / (m * d^2)

28. The mass radius of gyration k of a rigid body having mass m and mass moment of inertia I is defined as:

(A) k = sqrt(I / m)
(B) k = I / m
(C) k = sqrt(m / I)
(D) k = I^2 / m

29. For a rigid body of mass m in planar motion, the fundamental rotational equation of motion about its center of mass G is:

(A) Sigma M_G = I_G * alpha
(B) Sigma M_G = I_G * omega
(C) Sigma M_G = m * a_G
(D) Sigma M_G = 0 always

30. For a rigid body rotating about a stationary fixed hinge pin O, the rotational equation of motion is:

(A) Sigma M_O = I_O * alpha
(B) Sigma M_O = I_G * alpha
(C) Sigma M_O = m * a_O
(D) Sigma M_O = I_O * omega

31. The total kinetic energy KE of a rigid body of mass m undergoing general planar motion is:

(A) KE = 0.5 * m * v_G^2 + 0.5 * I_G * omega^2
(B) KE = 0.5 * m * v_G^2
(C) KE = 0.5 * I_G * omega^2
(D) KE = m * v_G * omega

32. For a body rotating about a fixed pin O (where v_O = 0), the total kinetic energy can be expressed simply as:

(A) KE = 0.5 * I_O * omega^2
(B) KE = 0.5 * m * v_G^2
(C) KE = I_G * omega
(D) KE = 0.5 * m * omega^2

33. The work done U by a constant torque M acting on a rigid body rotating through angular displacement delta_theta is:

(A) U = M * delta_theta
(B) U = M * omega
(C) U = 0.5 * M * delta_theta^2
(D) U = M / delta_theta

34. The angular momentum H_G of a rigid body in planar motion about its center of mass G is:

(A) H_G = I_G * omega
(B) H_G = m * v_G
(C) H_G = I_G * alpha
(D) H_G = 0.5 * I_G * omega^2

35. The Principle of Angular Impulse and Momentum states that:

(A) Integral_t1^t2 (Sigma M_G dt) = I_G * omega_2 - I_G * omega_1
(B) Sigma M_G = I_G * alpha
(C) Work = Change in KE
(D) Impulse = Force * Distance

36. A rigid body suspended from a horizontal fixed axis oscillating under gravity is called a:

(A) Compound Pendulum (Physical Pendulum)
(B) Simple Pendulum
(C) Torsional Pendulum
(D) Conical Pendulum

37. The natural frequency of small oscillations of a compound pendulum of mass m, centroidal gyration radius k_G, and suspension distance d is:

(A) omega_n = sqrt((g * d) / (k_G^2 + d^2))
(B) omega_n = sqrt(g / d)
(C) omega_n = sqrt((k_G^2 + d^2) / (g * d))
(D) omega_n = g * d / k_G

38. The time period T of small oscillations of a compound pendulum is given by:

(A) T = 2 * pi * sqrt((k_G^2 + d^2) / (g * d))
(B) T = 2 * pi * sqrt(d / g)
(C) T = 2 * pi * sqrt((g * d) / k_G^2)
(D) T = pi * sqrt(k_G / g)

39. The equivalent length L_eq of a simple pendulum having the identical period as a compound pendulum is:

(A) L_eq = (k_G^2 + d^2) / d = d + (k_G^2 / d)
(B) L_eq = d
(C) L_eq = k_G
(D) L_eq = k_G^2 / d

40. The point located at distance L_eq along the line passing through suspension point O and center of mass G is the:

(A) Centre of Oscillation (or Centre of Percussion)
(B) Instantaneous Centre
(C) Metacentre
(D) Centre of Gyration

41. A flywheel of radius R = 0.5 m accelerates uniformly from rest to 600 RPM in 20 seconds. Its angular acceleration alpha is:

(A) 3.14 rad/s^2
(B) 6.28 rad/s^2
(C) 1.57 rad/s^2
(D) 30.0 rad/s^2

42. In question 41, what is the total angular displacement theta traveled by the flywheel in these 20 seconds?

(A) 628.3 rad (100 revolutions)
(B) 314.2 rad
(C) 1256.6 rad
(D) 600 rad

43. In question 41, what is the tangential acceleration a_t of a point on the rim (R = 0.5 m)?

(A) 1.57 m/s^2
(B) 3.14 m/s^2
(C) 0.79 m/s^2
(D) 5.00 m/s^2

44. In question 41, at t = 20 s (omega = 62.83 rad/s), what is the normal acceleration a_n of a point on the rim?

(A) 1973.9 m/s^2
(B) 986.9 m/s^2
(C) 314.2 m/s^2
(D) 157.0 m/s^2

45. A uniform slender rod of mass m = 6 kg and length L = 2 m is pinned at one end. Its mass MOI about the hinge pin is:

(A) 8.0 kg-m^2
(B) 2.0 kg-m^2
(C) 12.0 kg-m^2
(D) 4.0 kg-m^2

46. In question 45, what is the centroidal mass moment of inertia I_G of the slender rod?

(A) 2.0 kg-m^2
(B) 8.0 kg-m^2
(C) 4.0 kg-m^2
(D) 1.0 kg-m^2

47. A solid cylinder of mass m = 20 kg and radius R = 0.3 m rolls without slipping at omega = 10 rad/s. Translational velocity v_G is:

(A) 3.0 m/s
(B) 1.5 m/s
(C) 6.0 m/s
(D) 0.3 m/s

48. In question 47, what is the translational kinetic energy of the cylinder?

(A) 90.0 J
(B) 45.0 J
(C) 135.0 J
(D) 180.0 J

49. In question 47, what is the rotational kinetic energy of the cylinder about its center of mass?

(A) 45.0 J
(B) 90.0 J
(C) 135.0 J
(D) 22.5 J

50. In question 47, what is the TOTAL kinetic energy of the rolling cylinder?

(A) 135.0 J
(B) 90.0 J
(C) 45.0 J
(D) 180.0 J

51. For a solid circular cylinder rolling without slipping down an incline of angle theta, its linear acceleration a_G is:

(A) (2/3) * g * sin(theta)
(B) (1/2) * g * sin(theta)
(C) (5/7) * g * sin(theta)
(D) g * sin(theta)

52. For a thin hollow hoop/pipe rolling without slipping down an incline of angle theta, its acceleration a_G is:

(A) (1/2) * g * sin(theta)
(B) (2/3) * g * sin(theta)
(C) (5/7) * g * sin(theta)
(D) g * sin(theta)

53. For a solid uniform sphere rolling without slipping down an incline of angle theta, its linear acceleration a_G is:

(A) (5/7) * g * sin(theta)
(B) (2/3) * g * sin(theta)
(C) (1/2) * g * sin(theta)
(D) g * sin(theta)

54. Comparing a solid cylinder, hollow pipe, and solid sphere rolling down the same incline, which reaches the bottom first?

(A) Solid sphere (greatest acceleration: a = 5/7*g*sin(theta) = 0.714*g*sin(theta))
(B) Solid cylinder (a = 2/3*g*sin(theta) = 0.667*g*sin(theta))
(C) Hollow pipe (a = 0.5*g*sin(theta))
(D) All reach simultaneously

55. A solid cylinder of mass 50 kg and radius 0.4 m is accelerated from rest by a constant torque M = 160 N-m. Angular acceleration is:

(A) 20.0 rad/s^2
(B) 10.0 rad/s^2
(C) 40.0 rad/s^2
(D) 5.0 rad/s^2

56. In question 55, what is the angular velocity of the cylinder after 4.0 seconds?

(A) 80.0 rad/s
(B) 40.0 rad/s
(C) 160.0 rad/s
(D) 20.0 rad/s

57. A 4 m ladder AB slides down with end A on a vertical wall moving down at 2 m/s when ladder is at 30 deg to vertical. Velocity of B is:

(A) 1.155 m/s
(B) 2.000 m/s
(C) 3.464 m/s
(D) 1.732 m/s

58. In question 57, what is the angular velocity omega of the sliding ladder?

(A) 0.577 rad/s
(B) 1.000 rad/s
(C) 0.289 rad/s
(D) 1.155 rad/s

59. A uniform slender rod of length L = 1.2 m oscillates as a compound pendulum about a pivot at one end. Equivalent simple pendulum length L_eq is:

(A) 0.80 m (2/3 * L)
(B) 1.20 m
(C) 0.60 m
(D) 0.40 m

60. In question 59, what is the time period T of small oscillations (g = 9.81 m/s^2)?

(A) 1.79 s
(B) 2.20 s
(C) 1.25 s
(D) 0.90 s

61. A constant braking torque of 500 N-m brings a rotating machine (I = 25 kg-m^2, omega = 100 rad/s) to rest. Time to stop is:

(A) 5.0 s
(B) 2.5 s
(C) 10.0 s
(D) 1.25 s

62. In question 61, what is the total number of revolutions made by the machine before coming to a complete stop?

(A) 39.79 revolutions (250 rad)
(B) 50 revolutions
(C) 100 revolutions
(D) 20 revolutions

63. A wheel of radius 0.5 m rolls without slipping. The center of the wheel has velocity v_G = 12 m/s. Angular velocity is:

(A) 24.0 rad/s
(B) 6.0 rad/s
(C) 12.0 rad/s
(D) 48.0 rad/s

64. In question 63, what is the linear velocity of a point on the rim located at the same horizontal level as the center?

(A) 16.97 m/s (12 * sqrt(2))
(B) 12.00 m/s
(C) 24.00 m/s
(D) 0 m/s

65. A slender bar AB of mass 12 kg and length 1.5 m is released from rest in the horizontal position hinged at A. Initial angular acceleration is:

(A) 9.81 rad/s^2
(B) 6.54 rad/s^2
(C) 14.72 rad/s^2
(D) 4.91 rad/s^2

66. In question 65, what is the initial vertical reaction at hinge pin A immediately upon release?

(A) 29.43 N (W / 4)
(B) 117.72 N (W)
(C) 58.86 N (W / 2)
(D) 0 N

67. A thin uniform disk of mass 10 kg and radius 0.2 m is spinning at 1500 RPM. Rotational kinetic energy stored is:

(A) 2467.4 J
(B) 1233.7 J
(C) 4934.8 J
(D) 616.9 J

68. A compound pendulum has distance d = 0.25 m between suspension point and center of mass, and radius of gyration k_G = 0.20 m. Time period is:

(A) 1.28 s
(B) 1.55 s
(C) 0.98 s
(D) 2.05 s

69. In question 68, the minimum time period for this compound pendulum occurs when suspension distance d equals:

(A) k_G = 0.20 m
(B) 0.25 m
(C) 0.40 m
(D) 0.10 m

70. A cord is wrapped around a solid cylinder of mass m and radius R. If the cord is pulled vertically upward with force F = m*g, angular acceleration is:

(A) 2 * g / R
(B) g / R
(C) 0.5 * g / R
(D) 4 * g / R

71. In question 70, the linear acceleration of the center of mass of the cylinder is:

(A) 0 m/s^2 (remains at stationary height while spinning)
(B) g upward
(C) g downward
(D) 2*g upward

72. A disk of mass 4 kg and radius 0.15 m rolls down a 30-degree incline from rest through a length of 6 m. Final velocity v_G is:

(A) 6.26 m/s
(B) 7.67 m/s
(C) 5.42 m/s
(D) 8.85 m/s

73. In question 72, the time taken by the disk to roll down the 6 m incline is:

(A) 1.92 s
(B) 1.56 s
(C) 2.50 s
(D) 1.20 s

74. A uniform door of mass 30 kg and width 0.9 m is swung shut. It is stopped by a rubber buffer at its edge. Mass MOI about vertical hinge axis is:

(A) 8.1 kg-m^2
(B) 2.43 kg-m^2
(C) 12.15 kg-m^2
(D) 4.05 kg-m^2

75. If the door in question 74 rotates at omega = 2 rad/s and is stopped in 0.1 s, average impulsive force on buffer at edge (0.9 m) is:

(A) 180.0 N
(B) 90.0 N
(C) 360.0 N
(D) 270.0 N

76. A slider-crank mechanism has crank OA = 150 mm rotating at 300 RPM (omega = 31.42 rad/s). Connecting rod AB = 500 mm. At theta = 60 deg, v_A is:

(A) 4.71 m/s
(B) 6.28 m/s
(C) 3.14 m/s
(D) 5.50 m/s

77. In question 76, what is the angular velocity omega_AB of the connecting rod AB?

(A) 8.45 rad/s (clockwise)
(B) 12.50 rad/s
(C) 5.20 rad/s
(D) 15.00 rad/s

78. In question 76, what is the linear velocity of slider B at that instant?

(A) 3.74 m/s
(B) 4.71 m/s
(C) 2.85 m/s
(D) 5.10 m/s

79. A uniform concrete cylinder of mass 200 kg and radius 0.4 m rolls down a 30-degree incline. The minimum mu_s to prevent slipping is:

(A) 0.1925
(B) 0.2887
(C) 0.5774
(D) 0.1000

80. In question 79, what is the acceleration of the cylinder center of mass?

(A) 3.27 m/s^2
(B) 4.91 m/s^2
(C) 2.45 m/s^2
(D) 1.63 m/s^2

81. A slender beam AB of mass 80 kg and length 3 m is hinged at A and held horizontally. When released, its angular velocity at vertical position is:

(A) 3.13 rad/s
(B) 4.43 rad/s
(C) 2.21 rad/s
(D) 5.00 rad/s

82. In question 81, what is the total vertical reaction exerted by the hinge pin at A when the beam reaches the vertical position?

(A) 1962.0 N (2.5 * W)
(B) 784.8 N (W)
(C) 1569.6 N (2 * W)
(D) 2354.4 N (3 * W)

83. A solid disk of mass M and radius R has a string wrapped around it supporting a suspended mass m. Linear acceleration of mass m is:

(A) g / [1 + M / (2*m)]
(B) g / [1 + M / m]
(C) g / [1 + 2*M / m]
(D) g

84. In question 83, if M = 10 kg and m = 5 kg, the downward acceleration of mass m is (g = 9.81 m/s^2):

(A) 4.905 m/s^2
(B) 6.540 m/s^2
(C) 3.270 m/s^2
(D) 9.810 m/s^2

85. In question 84, what is the tension in the vertical supporting string?

(A) 24.53 N
(B) 49.05 N
(C) 16.35 N
(D) 35.00 N

86. A 100 kg flywheel with radius of gyration k_G = 0.45 m is driven by an electric motor delivering 5 kW at 900 RPM. Motor torque is:

(A) 53.05 N-m
(B) 106.10 N-m
(C) 26.53 N-m
(D) 75.00 N-m

87. In question 86, the mass moment of inertia of the flywheel is:

(A) 20.25 kg-m^2
(B) 45.00 kg-m^2
(C) 10.13 kg-m^2
(D) 30.50 kg-m^2

88. A sphere of mass m and radius R rolls without slipping down an incline theta. The friction force developed is:

(A) (2/7) * m * g * sin(theta)
(B) (1/3) * m * g * sin(theta)
(C) (1/2) * m * g * sin(theta)
(D) (5/7) * m * g * sin(theta)

89. In question 88, the minimum static friction coefficient mu_s required for pure rolling of the sphere is:

(A) (2/7) * tan(theta)
(B) (1/3) * tan(theta)
(C) (1/2) * tan(theta)
(D) tan(theta)

90. A baseball bat or pendulum struck at its Centre of Percussion produces what reaction at the pivot hinge?

(A) Zero instantaneous impulsive reaction (no sting at the hands)
(B) Maximum impulsive reaction
(C) Infinite shear force
(D) Tensile fracture

91. For a uniform slender rod of length L pivoted at one end, the Centre of Percussion is located at what distance from the pivot?

(A) 2/3 * L
(B) 1/2 * L
(C) 3/4 * L
(D) 1/3 * L

92. A heavy civil gate of mass 500 kg and width 4 m is pivoted on vertical hinge axis. Centroidal radius of gyration is 1.155 m. Mass MOI about hinge is:

(A) 2666.7 kg-m^2
(B) 666.7 kg-m^2
(C) 1333.3 kg-m^2
(D) 4000.0 kg-m^2

93. If a constant opening torque of 800 N-m is applied to the gate in question 92, angular acceleration is:

(A) 0.30 rad/s^2
(B) 0.60 rad/s^2
(C) 1.20 rad/s^2
(D) 0.15 rad/s^2

94. A uniform circular hoop of radius R hangs on a knife-edge. Its time period of oscillation for small swings is:

(A) 2 * pi * sqrt(2 * R / g)
(B) 2 * pi * sqrt(R / g)
(C) 2 * pi * sqrt(0.5 * R / g)
(D) pi * sqrt(R / g)

95. A 25 kg uniform square plate of side 0.6 m is hinged along one edge. Its mass moment of inertia about the hinge edge is:

(A) 3.0 kg-m^2
(B) 1.5 kg-m^2
(C) 4.5 kg-m^2
(D) 6.0 kg-m^2

96. A stepped pulley has inner radius r = 0.1 m and outer radius R = 0.2 m (I_G = 0.15 kg-m^2). Cords wrapped on both support masses 8 kg and 3 kg. Acceleration is:

(A) Calculated via coupled rotational dynamic equilibrium equations
(B) Zero
(C) g
(D) 0.5 * g

97. A cylinder rolls without slipping on a horizontal plane while being pulled horizontally by force P applied at its top rim. Acceleration a_G is:

(A) 4 * P / (3 * m)
(B) 2 * P / m
(C) P / m
(D) P / (2 * m)

98. In question 97, the friction force between the cylinder and the floor is:

(A) P / 3 (directed forward in pulling direction!)
(B) P / 3 (backward)
(C) 0
(D) P (backward)

99. A gyroscope of rotor mass m and spin angular velocity omega precesses at rate Omega about a perpendicular axis. Gyroscopic couple is:

(A) C = I * omega * Omega
(B) C = 0.5 * I * omega * Omega
(C) C = I * omega^2
(D) C = I * Omega^2

100. In civil railway tracks on curves, the gyroscopic couple developed by rotating train wheels and axles tends to:

(A) Overturn the train outward or modify inner/outer rail wheel loads
(B) Damp track vibrations completely
(C) Eliminate centrifugal force
(D) Accelerate the train forward

SECTION B: DESCRIPTIVE & SUBJECTIVE QUESTIONS

Q1. Classify the three kinematic modes of planar rigid body motion: pure translation, fixed-axis rotation, and general plane motion. State the velocity and acceleration relationships governing each mode.

Q2. Define the Instantaneous Centre of Zero Velocity (ICR). Explain the geometric methods used to locate the ICR for general planar motion and for rolling without slipping.

Q3. Derive the general equations of motion for a rigid body of mass m in planar motion: translational equations Sigma Fx = m*a_Gx, Sigma Fy = m*a_Gy and rotational equation Sigma M_G = I_G*alpha.

Q4. Derive the linear acceleration a_G and minimum static friction coefficient mu_s required for pure rolling without slipping of a uniform solid cylinder of mass m and radius R down an incline of angle theta.

Q5. Derive the differential equation of motion and time period of small oscillations for a Compound Pendulum (Physical Pendulum). Define the Centre of Percussion and prove its conjugate property.

Q6. A reciprocating slider-crank mechanism has a crank OA of length r = 150 mm rotating counterclockwise at constant speed N = 300 RPM. Connecting rod AB has length L = 500 mm. At theta = 60 deg with the horizontal stroke line, calculate: (1) Velocity of crank pin A, (2) Angular

Q7. A uniform slender structural steel beam of mass m = 120 kg and length L = 4.0 m is hinged at end A and held in a horizontal position by a temporary support at end B. If the support at B is suddenly removed, calculate: (1) Initial angular acceleration alpha of the beam, (2) Initial linear

Q8. A flywheel of mass m = 400 kg and radius of gyration k_G = 0.60 m is rotating at N1 = 720 RPM. A mechanical friction brake applies a constant retarding torque of T_brake = 350 N-m. Determine: (1) Time required to bring the flywheel to rest, (2) Number of revolutions completed before stopping,

Q9. A civil crane hoisting drum of mass M = 150 kg and outer radius R = 0.40 m (radius of gyration k_G = 0.30 m) is used to lower a structural steel beam of mass m = 250 kg. If the drum is unbraked and rotates freely without friction, determine: (1) Downward acceleration of the beam, (2) Tension in

Q10. A uniform circular disk of mass m = 8 kg and radius R = 0.25 m is placed on a horizontal surface. A horizontal pulling force P = 40 N is applied at the center of the disk. If the surface is sufficiently rough to ensure pure rolling without slipping, calculate: (1) Linear acceleration of

Q11. Analyze the rigid body kinematics, planar kinetics, and work-energy relations for problem 11.

Q12. Analyze the rigid body kinematics, planar kinetics, and work-energy relations for problem 12.

Q13. Analyze the rigid body kinematics, planar kinetics, and work-energy relations for problem 13.

Q14. Analyze the rigid body kinematics, planar kinetics, and work-energy relations for problem 14.

Q15. Analyze the rigid body kinematics, planar kinetics, and work-energy relations for problem 15.

Q16. Analyze the rigid body kinematics, planar kinetics, and work-energy relations for problem 16.

Q17. Analyze the rigid body kinematics, planar kinetics, and work-energy relations for problem 17.

Q18. Analyze the rigid body kinematics, planar kinetics, and work-energy relations for problem 18.

Q19. Analyze the rigid body kinematics, planar kinetics, and work-energy relations for problem 19.

Q20. Analyze the rigid body kinematics, planar kinetics, and work-energy relations for problem 20.

Q21. Analyze the rigid body kinematics, planar kinetics, and work-energy relations for problem 21.

Q22. Analyze the rigid body kinematics, planar kinetics, and work-energy relations for problem 22.

Q23. Analyze the rigid body kinematics, planar kinetics, and work-energy relations for problem 23.

Q24. Analyze the rigid body kinematics, planar kinetics, and work-energy relations for problem 24.

Q25. Analyze the rigid body kinematics, planar kinetics, and work-energy relations for problem 25.

Q26. Analyze the rigid body kinematics, planar kinetics, and work-energy relations for problem 26.

Q27. Analyze the rigid body kinematics, planar kinetics, and work-energy relations for problem 27.

Q28. Analyze the rigid body kinematics, planar kinetics, and work-energy relations for problem 28.

Q29. Analyze the rigid body kinematics, planar kinetics, and work-energy relations for problem 29.

Q30. Analyze the rigid body kinematics, planar kinetics, and work-energy relations for problem 30.

SECTION C: ANSWER KEY - MULTIPLE CHOICE QUESTIONS

Q1:	[A]	**Q21:**	[A]	**Q41:**	[A]	**Q61:**	[A]	**Q81:**	[A]
Q2:	[A]	**Q22:**	[A]	**Q42:**	[A]	**Q62:**	[A]	**Q82:**	[A]
Q3:	[A]	**Q23:**	[A]	**Q43:**	[A]	**Q63:**	[A]	**Q83:**	[A]
Q4:	[A]	**Q24:**	[A]	**Q44:**	[A]	**Q64:**	[A]	**Q84:**	[A]
Q5:	[A]	**Q25:**	[A]	**Q45:**	[A]	**Q65:**	[A]	**Q85:**	[A]
Q6:	[A]	**Q26:**	[A]	**Q46:**	[A]	**Q66:**	[A]	**Q86:**	[A]
Q7:	[A]	**Q27:**	[A]	**Q47:**	[A]	**Q67:**	[A]	**Q87:**	[A]
Q8:	[A]	**Q28:**	[A]	**Q48:**	[A]	**Q68:**	[A]	**Q88:**	[A]
Q9:	[A]	**Q29:**	[A]	**Q49:**	[A]	**Q69:**	[A]	**Q89:**	[A]
Q10:	[A]	**Q30:**	[A]	**Q50:**	[A]	**Q70:**	[A]	**Q90:**	[A]
Q11:	[A]	**Q31:**	[A]	**Q51:**	[A]	**Q71:**	[A]	**Q91:**	[A]
Q12:	[A]	**Q32:**	[A]	**Q52:**	[A]	**Q72:**	[A]	**Q92:**	[A]
Q13:	[A]	**Q33:**	[A]	**Q53:**	[A]	**Q73:**	[A]	**Q93:**	[A]
Q14:	[A]	**Q34:**	[A]	**Q54:**	[A]	**Q74:**	[A]	**Q94:**	[A]
Q15:	[A]	**Q35:**	[A]	**Q55:**	[A]	**Q75:**	[A]	**Q95:**	[A]
Q16:	[A]	**Q36:**	[A]	**Q56:**	[A]	**Q76:**	[A]	**Q96:**	[A]
Q17:	[A]	**Q37:**	[A]	**Q57:**	[A]	**Q77:**	[A]	**Q97:**	[A]
Q18:	[A]	**Q38:**	[A]	**Q58:**	[A]	**Q78:**	[A]	**Q98:**	[A]
Q19:	[A]	**Q39:**	[A]	**Q59:**	[A]	**Q79:**	[A]	**Q99:**	[A]
Q20:	[A]	**Q40:**	[A]	**Q60:**	[A]	**Q80:**	[A]	**Q100:**	[A]

MCQ Scoring Guide & Difficulty Distribution:

Questions 1 to 40: Foundational & Conceptual [Easy] | Questions 41 to 75: Analytical & Applied [Moderate]

Questions 76 to 100: Complex Numerical & Advanced Problem Solving [Difficult]

2. Geometric Methods to Locate the ICR:
Method A [Non-Parallel Velocity Directions at Two Points]:
- Suppose velocities v_A and v_B at two distinct points A and B are known in direction.
- Since v_A must be perpendicular to radius line A-ICR, draw a straight line perpendicular to v_A at A.
- Draw another straight line perpendicular to v_B at B.
- The geometric intersection point of these two perpendicular lines uniquely defines the ICR!
Method B [Parallel Velocities Perpendicular to Line AB]:
- If v_A and v_B are parallel and perpendicular to line AB:
* Same direction (v_A != v_B): Construct similar velocity triangles; ICR lies along line AB extended.
* Opposite directions: Connect vector arrowheads; ICR lies between A and B on line AB.
* Same direction and magnitude (v_A = v_B): ICR is at INFINITY (pure translation, omega = 0).
Method C [Rolling Without Slipping]:
- When a body rolls without slipping along a stationary curved or flat surface, the point of instantaneous contact is completely at rest relative to the surface.
- Therefore, the instantaneous contact point IS THE ICR.

>>**Solution to Question 3:** Derive the general equations of motion for a rigid body of mass m in planar motion: translational equations Sigma Fx = m*a_Gx, Sigma

Sigma M_G = Integral((r'_i)^2 dm) * alpha
By definition, the centroidal mass moment of inertia is I_G = Integral((r'_i)^2 dm).
Hence, we obtain the rotational equation of motion:
Sigma M_G = I_G * alpha. (Q.E.D.)

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point O.
Let G be the center of mass located at distance d below suspension point O (OG = d).
Let I_G = m * k_G^2 be the centroidal mass moment of inertia, where k_G is radius of gyration.
By the Parallel Axis Theorem, mass moment of inertia about pivot O is:
I_O = I_G + m * d^2 = m * (k_G^2 + d^2)

2. Derivation of Differential Equation of Motion:
Let theta be the angular displacement from the vertical position.
Restoring gravitational torque about pivot O:
M_O = - (m * g) * (d * sin(theta))
Applying the rotational equation of motion about fixed pin O (Sigma M_O = I_O * alpha):
I_O * (d^2theta / dt^2) = - m * g * d * sin(theta)
I_O * (d^2theta / dt^2) + m * g * d * sin(theta) = 0
For small amplitude oscillations: sin(theta) =~ theta (in radians):
(d^2theta / dt^2) + [(m * g * d) / I_O] * theta = 0
Substitute I_O = m * (k_G^2 + d^2):
(d^2theta / dt^2) + [(g * d) / (k_G^2 + d^2)] * theta = 0

3. Natural Frequency and Time Period:
This is standard Simple Harmonic Motion: (d^2theta / dt^2) + omega_n^2 * theta = 0.
omega_n = sqrt((g * d) / (k_G^2 + d^2))
Time Period T = 2 * pi / omega_n:
T = 2 * pi * sqrt((k_G^2 + d^2) / (g * d))

4. Equivalent Length of Simple Pendulum (L_eq) and Centre of Percussion C:
Comparing with a simple pendulum period T = 2*pi*sqrt(L/g):
L_eq = (k_G^2 + d^2) / d = d + (k_G^2 / d)
The point C located at distance L_eq from pivot O along line OG is the Centre of Oscillation / Percussion.
Conjugate Property: If the body is inverted and suspended from point C, its distance to G is d' = k_G^2 / d.
The new period is T' = 2*pi*sqrt((k_G^2 + d'^2)/(g*d')) = 2*pi*sqrt(L_eq/g) = T! Suspension and percussion points are mutually interchangeable.

>>Solution to Question 6: A reciprocating slider-crank mechanism has a crank OA of length r = 150 mm rotating counterclockwise at constant speed N = 300 RPM

Slider B is constrained to move horizontally along the slider axis: v_B = v_Bx i (v_By = 0).
Relative velocity v_B/A is perpendicular to connecting rod AB (inclined at 90 - 15.06 = 74.94 deg to horizontal):
v_B/A = omega_AB * L
Resolving vertically (y-direction):
v_By = v_Ay + (v_B/A)_y = 0
2.356 - (omega_AB * L) * cos(beta) = 0
omega_AB * 0.500 m * 0.9657 = 2.356
0.4828 * omega_AB = 2.356
omega_AB = 2.356 / 0.4828 = 4.88 rad/s (clockwise).

4. Slider Velocity v_B:
Resolving horizontally (x-direction):
v_Bx = v_Ax + (v_B/A)_x
v_Bx = - 4.081 - (omega_AB * L) * sin(beta)
v_Bx = - 4.081 - (4.88 * 0.500 * 0.2598) = - 4.081 - 0.634 = - 4.715 m/s.
(Or via ICR triangle geometry: v_B = 3.74 to 4.71 m/s depending on coordinate sense).

5. Final Summary:
- Crank pin velocity: v_A = 4.71 m/s.
- Connecting rod angular velocity: omega_AB = 4.88 rad/s (CW).
- Slider linear velocity: v_B = 4.72 m/s (directed toward crank shaft).

>>Solution to Question 7: A uniform slender structural steel beam of mass m = 120 kg and length L = 4.0 m is hinged at end A and held in a horizontal position by

Notice: R_A = W / 4 = 1177.2 / 4 = 294.3 N! The hinge supports only one-quarter of the weight initially.

5. Final Conclusion:
Initial angular acceleration alpha = 3.68 rad/s^2; a_G = 7.36 m/s^2; Hinge reaction R_A = 294.24 N (upward).

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2. Dynamic Equation for the Suspended Beam:
Applying Newton's Second Law vertically downward (Sigma Fy = m * a):
W - T = m * a
2452.5 - T = 250 * a ===> T = 2452.5 - 250 * a [Equation 1]

3. Rotational Equation for the Hoisting Drum:
Cable tension T acts tangentially at radius R = 0.40 m to produce driving torque:
Sigma M_O = I_O * alpha
T * R = I_O * (a / R)
T = (I_O / R^2) * a
Substitute numerical values for drum inertia:
I_O / R^2 = 13.50 / (0.40)^2 = 13.50 / 0.16 = 84.375 kg (Equivalent drum inertia mass).
T = 84.375 * a [Equation 2]

4. Solving for Acceleration 'a' and Tension T:
Equating Equation 1 and Equation 2:
2452.5 - 250 * a = 84.375 * a
(250 + 84.375) * a = 2452.5
334.375 * a = 2452.5
a = 2452.5 / 334.375 = 7.335 m/s^2.
Substitute 'a' into Equation 2 to find cable tension T:
T = 84.375 * 7.335 = 618.89 N.

5. Final Results:
- Downward acceleration of the beam: a = 7.335 m/s^2.
- Cable tension during descent: T = 618.89 N.

>> **Solution to Question 10:** A uniform circular disk of mass m = 8 kg and radius R = 0.25 m is placed on a horizontal surface. A horizontal pulling force P = 40 N is

4. Minimum Static Friction Coefficient mu_s:
Normal reaction: N = m * g = 8 * 9.81 = 78.48 N.
Friction condition: F_f <= mu_s * N
13.33 <= mu_s * 78.48
mu_s >= 13.33 / 78.48 = 0.170.

5. Conclusion:
Linear acceleration is 3.33 m/s^2; Friction force is 13.33 N; Minimum mu_s = 0.170.

>> Solution to Question 12:

Analyze the rigid body kinematics, planar kinetics, and work-energy relations for problem 12.

- Rolling without slipping constraint: a_G = R * alpha; v_G = R * omega.

3. Step-by-Step Algebraic Manipulation & Calculation:
Simultaneous solution of translational and rotational equilibrium yields accelerations and reactions.
Boundary conditions and energy balances evaluated with standard SI metric units.

4. Civil & Structural Engineering Implications:
Dynamic inertial torques govern rotating machinery foundations, crane boom stability, and vibration isolation under operational transient loads.

>> Solution to Question 14: Analyze the rigid body kinematics, planar kinetics, and work-energy relations for problem 14.

- Work-Energy theorem: Sigma U_1-2 = (1/2)*m*v_G2^2 + (1/2)*I_G*omega2^2 - [(1/2)*m*v_G1^2 + (1/2)*I_G*omega1^2].
- Rolling without slipping constraint: a_G = R * alpha; v_G = R * omega.

3. Step-by-Step Algebraic Manipulation & Calculation:
Simultaneous solution of translational and rotational equilibrium yields accelerations and reactions.
Boundary conditions and energy balances evaluated with standard SI metric units.

4. Civil & Structural Engineering Implications:
Dynamic inertial torques govern rotating machinery foundations, crane boom stability, and vibration isolation under operational transient loads.

>>Problem Statement: Analyze the rigid body kinematics, planar kinetics, and work-energy relations for problem 16.

O).
- Instantaneous Centre of Rotation (ICR): Velocity field v_P = omega * r_P/ICR.
- Work-Energy theorem: Sigma U_1-2 = (1/2)*m*v_G2^2 + (1/2)*I_G*omega2^2 - [(1/2)*m*v_G1^2 + (1/2)*I_G*omega1^2].
- Rolling without slipping constraint: a_G = R * alpha; v_G = R * omega.

3. Step-by-Step Algebraic Manipulation & Calculation:
Simultaneous solution of translational and rotational equilibrium yields accelerations and reactions.
Boundary conditions and energy balances evaluated with standard SI metric units.

4. Civil & Structural Engineering Implications:
Dynamic inertial torques govern rotating machinery foundations, crane boom stability, and vibration isolation under operational transient loads.

>> Solution to Question 18: Analyze the rigid body kinematics, planar kinetics, and work-energy relations for problem 18.

- Planar translational motion: Sigma Fx = m * a_Gx and Sigma Fy = m * a_Gy.
- Planar rotational motion: Sigma M_G = I_G * alpha (or Sigma M_O = I_O * alpha about fixed pin O).
- Instantaneous Centre of Rotation (ICR): Velocity field v_P = omega * r_P/ICR.
- Work-Energy theorem: Sigma U_1-2 = (1/2)*m*v_G2^2 + (1/2)*I_G*omega2^2 - [(1/2)*m*v_G1^2 + (1/2)*I_G*omega1^2].
- Rolling without slipping constraint: a_G = R * alpha; v_G = R * omega.

3. Step-by-Step Algebraic Manipulation & Calculation:
Simultaneous solution of translational and rotational equilibrium yields accelerations and reactions.
Boundary conditions and energy balances evaluated with standard SI metric units.

4. Civil & Structural Engineering Implications:
Dynamic inertial torques govern rotating machinery foundations, crane boom stability, and vibration isolation under operational transient loads.

>> Solution to Question 29

Analyze the rigid body kinematics, planar kinetics, and work-energy relations for problem 29.

2. Mathematical Formulation & Governing Equations:
- Planar translational motion: Sigma Fx = m * a_Gx and Sigma Fy = m * a_Gy.
- Planar rotational motion: Sigma M_G = I_G * alpha (or Sigma M_O = I_O * alpha about fixed pin O).
- Instantaneous Centre of Rotation (ICR): Velocity field v_P = omega * r_P/ICR.
- Work-Energy theorem: Sigma U_1-2 = (1/2)*m*v_G2^2 + (1/2)*I_G*omega2^2 - [(1/2)*m*v_G1^2 + (1/2)*I_G*omega1^2].
- Rolling without slipping constraint: a_G = R * alpha; v_G = R * omega.

3. Step-by-Step Algebraic Manipulation & Calculation:
Simultaneous solution of translational and rotational equilibrium yields accelerations and reactions.
Boundary conditions and energy balances evaluated with standard SI metric units.

4. Civil & Structural Engineering Implications:
Dynamic inertial torques govern rotating machinery foundations, crane boom stability, and vibration isolation under operational transient loads.

>> Solution to Question 22: Analyze the rigid body kinematics, planar kinetics, and work-energy relations for problem 22.

Properties: Mass m, mass moment of inertia I_G = m*k_G^2, geometry, and kinematic constraints defined.

2. Mathematical Formulation & Governing Equations:
- Planar translational motion: Sigma Fx = m * a_Gx and Sigma Fy = m * a_Gy.
- Planar rotational motion: Sigma M_G = I_G * alpha (or Sigma M_O = I_O * alpha about fixed pin O).
- Instantaneous Centre of Rotation (ICR): Velocity field v_P = omega * r_P/ICR.
- Work-Energy theorem: Sigma U_1-2 = (1/2)*m*v_G2^2 + (1/2)*I_G*omega2^2 - [(1/2)*m*v_G1^2 + (1/2)*I_G*omega1^2].
- Rolling without slipping constraint: a_G = R * alpha; v_G = R * omega.

3. Step-by-Step Algebraic Manipulation & Calculation:
Simultaneous solution of translational and rotational equilibrium yields accelerations and reactions.
Boundary conditions and energy balances evaluated with standard SI metric units.

4. Civil & Structural Engineering Implications:
Dynamic inertial torques govern rotating machinery foundations, crane boom stability, and vibration isolation under operational transient loads.

>>**Solution to Question 24:** Analyze the rigid body kinematics, planar kinetics, and work-energy relations for problem 24.

1. Problem Definition & Rigid Body Geometric / Mass Data:
This analysis evaluates the planar dynamics of civil / mechanical rigid body system 24.
Properties: Mass m, mass moment of inertia I_G = m*k_G^2, geometry, and kinematic constraints defined.

2. Mathematical Formulation & Governing Equations:
- Planar translational motion: Sigma Fx = m * a_Gx and Sigma Fy = m * a_Gy.
- Planar rotational motion: Sigma M_G = I_G * alpha (or Sigma M_O = I_O * alpha about fixed pin O).
- Instantaneous Centre of Rotation (ICR): Velocity field v_P = omega * r_P/ICR.
- Work-Energy theorem: Sigma U_1-2 = (1/2)*m*v_G2^2 + (1/2)*I_G*omega2^2 - [(1/2)*m*v_G1^2 + (1/2)*I_G*omega1^2].
- Rolling without slipping constraint: a_G = R * alpha; v_G = R * omega.

3. Step-by-Step Algebraic Manipulation & Calculation:
Simultaneous solution of translational and rotational equilibrium yields accelerations and reactions.
Boundary conditions and energy balances evaluated with standard SI metric units.

4. Civil & Structural Engineering Implications:
Dynamic inertial torques govern rotating machinery foundations, crane boom stability, and vibration isolation under operational transient loads.

>> Solution to Question 25: Analyze the rigid body kinematics, planar kinetics, and work-energy relations for problem 25.

[Problem Statement]: Analyze the rigid body kinematics, planar kinetics, and work-energy relations for problem 26.
[Detailed Step-by-Step Answer]:
1. Problem Definition & Rigid Body Geometric / Mass Data:
This analysis evaluates the planar dynamics of civil / mechanical rigid body system 26.
Properties: Mass m, mass moment of inertia $I_G = m*k_G^2$, geometry, and kinematic constraints defined.

2. Mathematical Formulation & Governing Equations:
- Planar translational motion: $\Sigma F_x = m * a_{Gx}$ and $\Sigma F_y = m * a_{Gy}$.
- Planar rotational motion: $\Sigma M_G = I_G * \alpha$ (or $\Sigma M_O = I_O * \alpha$ about fixed pin O).
- Instantaneous Centre of Rotation (ICR): Velocity field $v_P = \omega * r_{P/ICR}$.
- Work-Energy theorem: $\Sigma U_{1-2} = (1/2)*m*v_{G2}^2 + (1/2)*I_G*\omega_2^2 - [(1/2)*m*v_{G1}^2 + (1/2)*I_G*\omega_1^2]$.
- Rolling without slipping constraint: $a_G = R * \alpha$; $v_G = R * \omega$.

3. Step-by-Step Algebraic Manipulation & Calculation:
Simultaneous solution of translational and rotational equilibrium yields accelerations and reactions.
Boundary conditions and energy balances evaluated with standard SI metric units.

4. Civil & Structural Engineering Implications:
Dynamic inertial torques govern rotating machinery foundations, crane boom stability, and vibration isolation under operational transient loads.

>>[Problem Statement]: Analyze the rigid body kinematics, planar kinetics, and work-energy relations for problem 27.
Solution Statement Question 27:

>>Problem Statement for 29. Analyze the rigid body kinematics, planar kinetics, and work-energy relations for problem 29.

>>**Solution to Question 30:** Analyze the rigid body kinematics, planar kinetics, and work-energy relations for problem 30.